

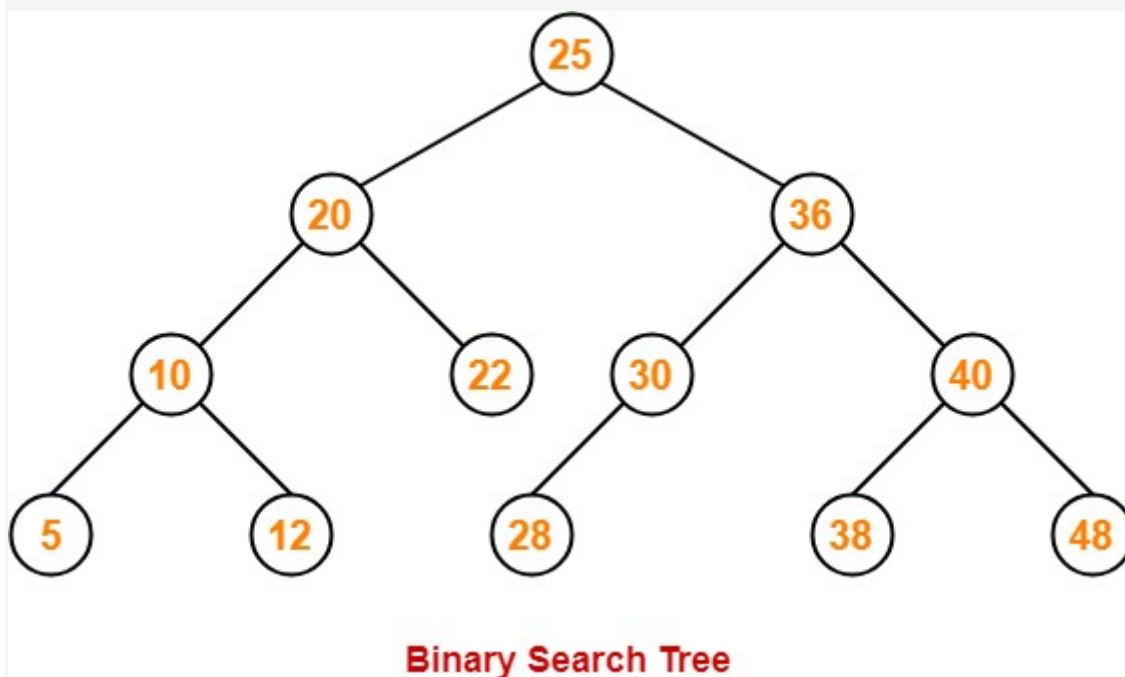
Binary Search Tree | Example | Construction

Binary Search Tree is a special kind of binary tree in which nodes are arranged in a specific order.

In a binary search tree (BST), each node contains-

- Only smaller values in its left sub tree
- Only larger values in its right sub tree

Example-



Number of Binary Search Trees-

$$\text{Number of distinct Binary Search Trees possible with } n \text{ distinct keys} = \frac{{}^{2n}C_n}{n+1}$$

Example-

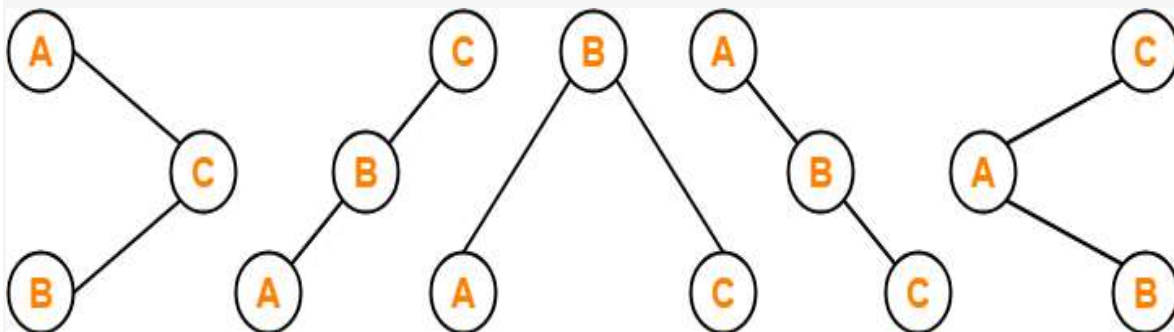
Number of distinct binary search trees possible with 3 distinct keys

$$= {}^{2 \times 3}C_3 / 3+1$$

$$= {}^6C_3 / 4$$

$$= 5$$

If three distinct keys are A, B and C, then 5 distinct binary search trees are-



Binary Search Tree Construction-

Let us understand the construction of a binary search tree using the following example-

Example-

Construct a Binary Search Tree (BST) for the following sequence of numbers-

50, 70, 60, 20, 90, 10, 40, 100

When elements are given in a sequence,

- Always consider the first element as the root node.
- Consider the given elements and insert them in the BST one by one.

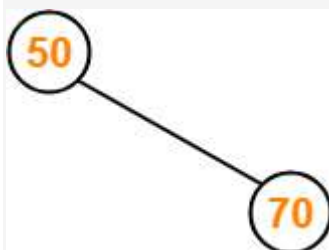
The binary search tree will be constructed as explained below-

Insert 50-



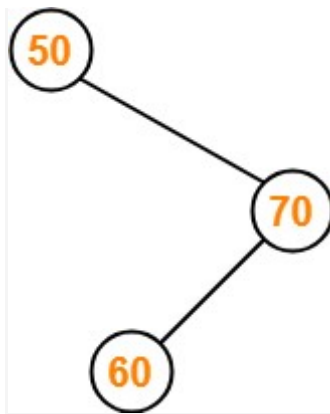
Insert 70-

- As $70 > 50$, so insert 70 to the right of 50.



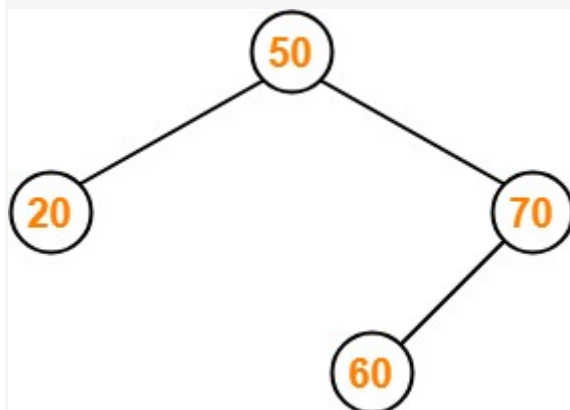
Insert 60-

- As $60 > 50$, so insert 60 to the right of 50.
- As $60 < 70$, so insert 60 to the left of 70.



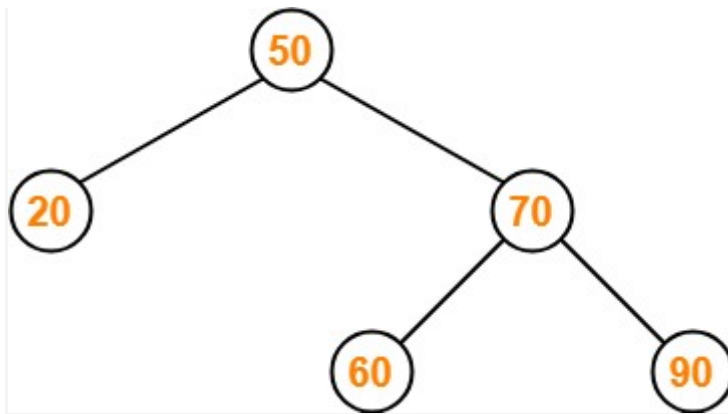
Insert 20-

- As $20 < 50$, so insert 20 to the left of 50.



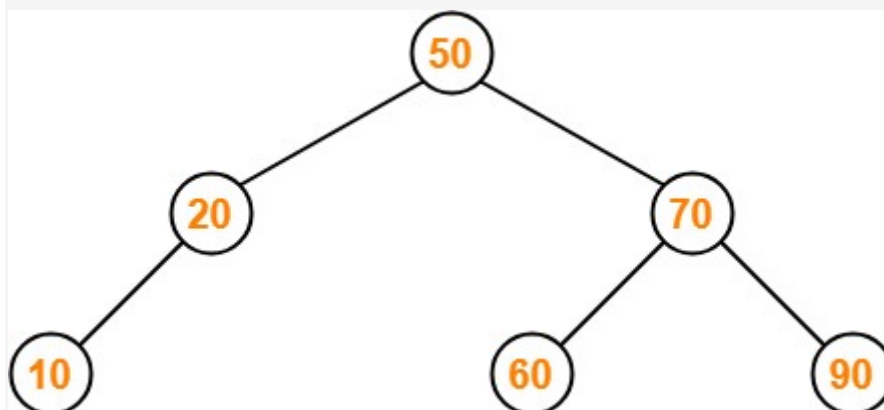
Insert 90-

- As $90 > 50$, so insert 90 to the right of 50.
- As $90 > 70$, so insert 90 to the right of 70.



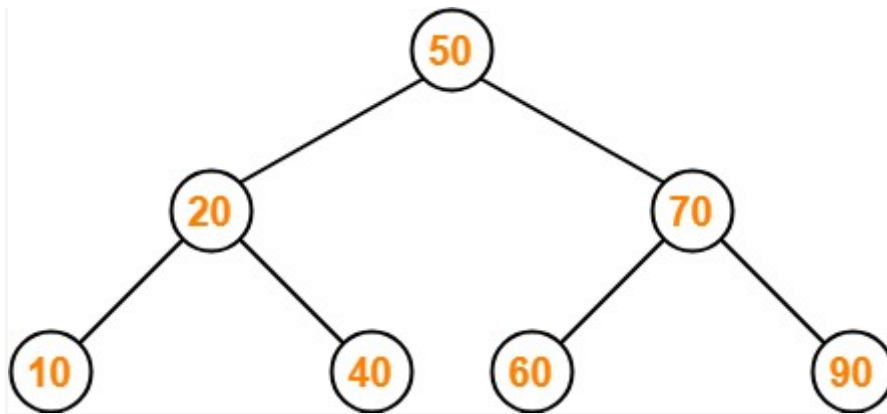
Insert 10-

- As $10 < 50$, so insert 10 to the left of 50.
- As $10 < 20$, so insert 10 to the left of 20.



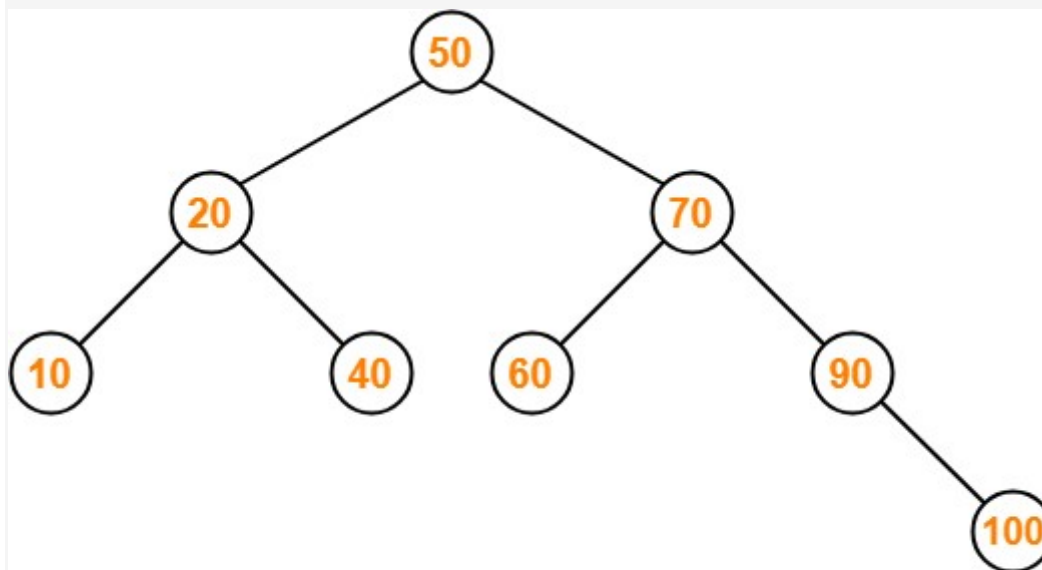
Insert 40-

- As $40 < 50$, so insert 40 to the left of 50.
- As $40 > 20$, so insert 40 to the right of 20.



Insert 100-

- As $100 > 50$, so insert 100 to the right of 50.
- As $100 > 70$, so insert 100 to the right of 70.
- As $100 > 90$, so insert 100 to the right of 90.



Binary Search Tree

This is the required Binary Search Tree.

To gain better understanding about Binary Search Trees,

[Watch this Video Lecture](#)

PRACTICE PROBLEMS BASED ON BINARY SEARCH TREES-

Problem-01:

A binary search tree is generated by inserting in order of the following integers-

50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24

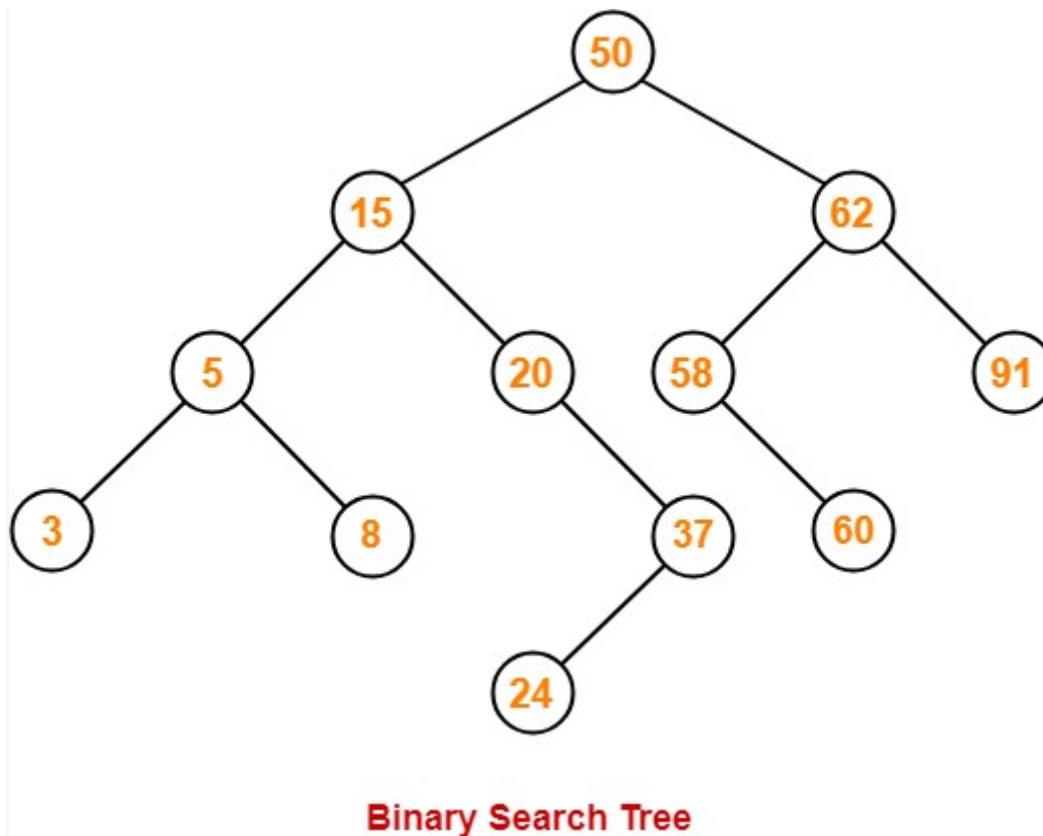
The number of nodes in the left subtree and right subtree of the root respectively is _____.

- A. (4, 7)
- B. (7, 4)
- C. (8, 3)
- D. (3, 8)

Solution-

Using the above discussed steps, we will construct the binary search tree.

The resultant binary search tree will be-



Clearly,

- Number of nodes in the left subtree of the root = 7
- Number of nodes in the right subtree of the root = 4

Thus, Option (B) is correct.

Problem-02:

How many distinct binary search trees can be constructed out of 4 distinct keys?

- A. 5
- B. 14
- C. 24
- D. 35

Solution-

Number of distinct binary search trees possible with 4 distinct keys

$$= {}^{2n}C_n / n+1$$

$$= {}^{2 \times 4}C_4 / 4+1$$

$$= {}^8C_4 / 5$$

$$= 14$$

Thus, Option (B) is correct.

Problem-03:

The numbers 1, 2, ..., n are inserted in a binary search tree in some order. In the resulting tree, the right subtree of the root contains p nodes. The first number to be inserted in the tree must be-

- A. p
- B. p+1
- C. n-p
- D. n-p+1

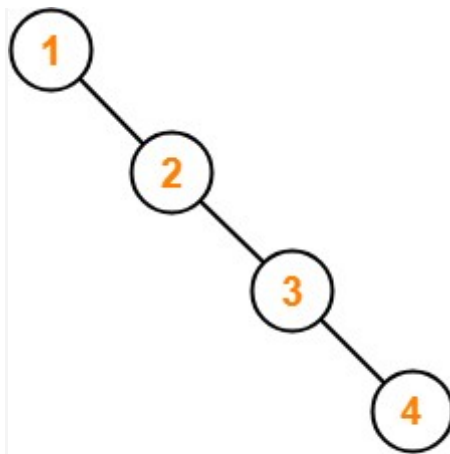
Solution-

Let n = 4 and p = 3.

Then, given options reduce to-

- A. 3
- B. 4
- C. 1
- D. 2

Our binary search tree will be as shown-



Binary Search Tree

Clearly, first inserted number = 1.

Thus, Option (C) is correct.

Problem-04:

We are given a set of n distinct elements and an unlabeled binary tree with n nodes. In how many ways can we populate the tree with given set so that it becomes a binary search tree?

- A. 0
- B. 1
- C. $n!$
- D. $C(2n, n) / n+1$

Solution-

Option (B) is correct.